

Fig. 4: Equivalent circuit of an antenna

Advantages of harvested RF energy

- Available on demand
- Works in perpetually dark locations
- Works in hazardous location
- Provides mobility
- Provides tracking capability
- Can take advantage of electricity tariffs
- Can change a secondary battery

Along similar lines, a slight change in the matching circuit parameters significantly alters the frequency range in which the efficiency of energy conversion is maximum, often by several MHz. Hence, RF-harvesting circuits involve a complex interplay of design choices, which must be considered together. This problem is addressed by considering a multi-stage design of the voltage multiplier, whose operating points are decided by solving an optimisation framework.

Antenna. An antenna is a metallic device (like a rod or wire) for radiating or receiving radio waves. It is a means of transmitting and receiving radio waves. In other words, an antenna is a transition structure between free space and a guided device. The guided device or transmission line may take the form of a coaxial line or a hollow pipe (waveguide), and is used to transport electromagnetic energy from the transmitting source to or from the antenna to the receiver.

Communication devices generally have omni-directional antennae that propagate RF energy in most directions. This maximises connectivity for mobile applications. Energy transmitted from the wireless sources is much higher, up to 30W for 10GHz frequency, but only a small amount can be scavenged in

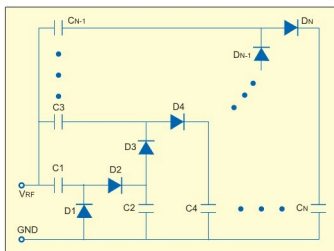


Fig. 5: Charge-pump rectifier

the real environment. The rest is dissipated as heat or absorbed by other materials.

An antenna is required for harvesting RF energy. In an RF energy-harvesting system, the antenna (as receiver) intercepts electromagnetic waves passing by and converts these into electrical signals. A typical antenna can be modelled as AC voltage source series impedance, as shown in Fig. 4.

P_{RF} is power that is received by an antenna; R_s is radiation resistance, which stands for the power used for receiving electromagnetic waves; R_{loss} is loss resistance, representing the actual resistance including the materials of the antenna and dielectric loss and X_{ant} can either be inductive or capacitive depending on the specific antenna.

There are a number of parameters that antennae designers have to pay attention to. Power gain is the ratio of the maximum power density that an antenna can radiate from a certain angle at an arbitrary distance to power density at the same distance of a hypothetical isotropic antenna. It can be thought of as actual maximum power density over the ideal (loss-less) average power density. The effective antenna aperture is closely related to the available power an antenna can deliver. Some other key parameters include path loss, polarisation, efficiency, directivity and so on.

Rectenna. The device used to harvest RF energy is called a rectenna. It refers to a rectifying antenna. A rectenna converts RF energy into DC energy. A typical rectenna has an antenna, a matching circuit and a rectifier.

A rectenna is a hybrid of an antenna and Schottky diode placed at an antenna's feed point. It directly

converts RF to DC signal. Different topologies are available to work as components of the rectenna. Different topologies have been developed for RF-DC rectifier design, such as charge-pump, differential drive bridge rectifier and gate cross-connected differential-drive bridge rectifier.

Charge-pump rectifier. A charge-pump rectifier, also known as voltage multiplier circuit, as shown in Fig. 4, has been widely used in micro power-harvesting circuits. The basic structure of the conventionally-used charge-pump rectifier was proposed in 1976 and originally used as a DC-DC up-converter.

In the RF-DC rectifier application, the circuit is adapted by using RF input as one of the complementary clock signals, and grounding the other clock line and DC input line. To understand the operating principle of the rectifier, first two stages of the multiplier are considered, which are also usually denoted as voltage-doubler.

The operation can be analysed in two cycles: negative half-cycle, where input RF signal is negative, and positive half-cycle, where input RF signal is positive.

Assume, threshold voltage of the diode is V_T and amplitude of input RF signal is V_{RF} . In the first negative half-cycle, D1 is conducted and charges are transferred to the right-end plate of C1. At the end of